International Baccalaureate®
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MARKSCHEME

SPECIMEN PAPER

PHYSICS

Higher Level

Paper 2

11 pages

General Marking Instructions

Subject Details: Physics HL Paper 2 Markscheme

Mark Allocation

Candidates are required to answer **ALL** questions. Maximum total = **[95 marks]**.

Markscheme format example:

Question			Answers	Notes	Total
4.	**b**	**ii**	the displacement and acceleration ✓ are in opposite directions ✓	*Accept **force** for **acceleration**.*	**2**

1. Each row in the "Question" column relates to the smallest subpart of the question.

2. The maximum mark for each question subpart is indicated in the "Total" column.

3. Each marking point in the "Answers" column is shown by means of a tick (✓) at the end of the marking point.

4. A question subpart may have more marking points than the total allows. This will be indicated by "**max**" written after the mark in the "Total" column. The related rubric, if necessary, will be outlined in the "Notes" column.

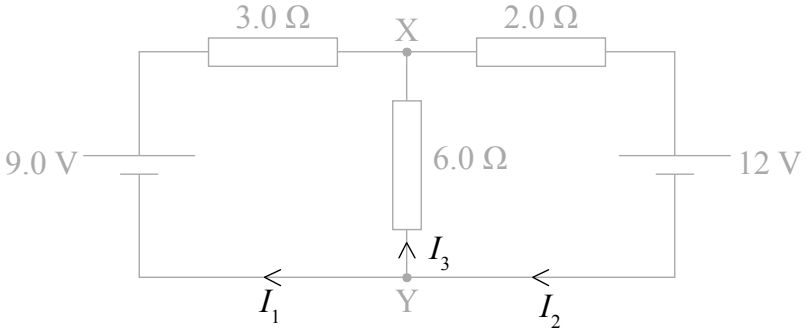
5. An alternative wording is indicated in the "Answers" column by a slash (/). Either wording can be accepted.

6. An alternative answer is indicated in the "Answers" column by "***OR***" on the line between the alternatives. Either answer can be accepted.

7. Words in angled brackets ❮ ❯ in the "Answers" column are not necessary to gain the mark.

8. Words that are <u>underlined</u> are essential for the mark.

Question			Answers	Notes	Total
1.	**a**		use of suitable kinematic equation *eg*: $-48 = 48 - 9.81t$ ✓ 9.8 ‹s› ✓	*Award **[2]** for a bald correct answer.*	**2**
	b		$0 = 48^2 - 2a8$ so $a = 144$ ‹$\mathrm{m\,s^{-2}}$› ✓ $F_{\text{net}} = 480 \times 144$ ‹$= 6.9 \times 10^4$› ✓ average force ‹$= 6.9 \times 10^4 + 0.47 \times 10^4$› $= 7.4 \times 10^4$ ‹N› ✓		**3**
	c		reaction force is zero ✓ because object and container fall at same rate ✓		**2**
	d	**i**	volume $= 120 \times \pi \times (3.5)^2 = 4620$ ‹$\mathrm{m^3}$› ✓ $n = \dfrac{0.96 \times \text{volume}}{(8.31 \times 300)} = 1.78$ ✓ number of molecules $= 6.02 \times 10^{23} \times n = 1.1 \times 10^{24}$ ✓		**3**
	d	**ii**	‹yes› because pressure is low ✓ and temperature is high/moderate ✓		**2**
	e	**i**	stopping distance ‹in polystyrene›/8 m ✓	*Do not accept distance unqualified.*	**1**
	e	**ii**	gradient decreases as time increases before hitting the polystyrene ✓ lower maximum ✓ graph must go on longer before deceleration ✓ same total area by eye ✓	*Accept a graph reaching terminal speed.*	**3 max**

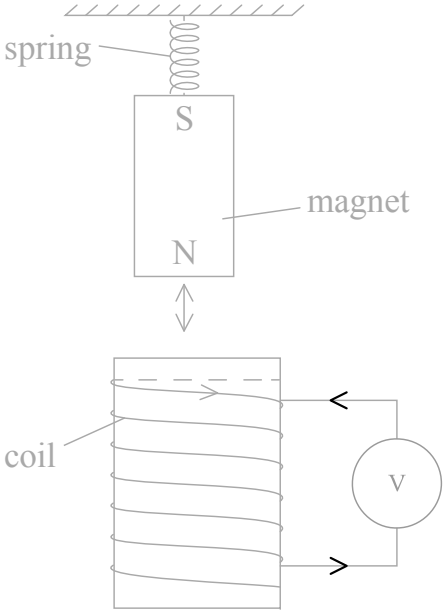
Question			Answers	Notes	Total
2.	**a**		Kirchhoff 1 applied correctly ✓ Kirchhoff 2 applied to one loop correctly ✓ Kirchhoff 2 applied to a second loop correctly ✓ simultaneous equations constructed correctly to give answer ✓ direction of current stated towards X ✓  *eg*: $I_2 = I_1 + I_3$ $9 = 3I_1 - 6I_3$ $12 = -2I_2 - 6I_3$ $9 = 3I_2 - 3I_3 - 6I_3$ $36 = -6I_2 - 18I_3$ $18 = 6I_2 - 18I_3$ $54 = -36I_3$ $I_3 = -1.5\,\text{A}$	*Allow ECF from incorrect sign in the last marking point.* *Negative sign means current flow is from X to Y.*	**4 max**
	b	**i**	potential difference = ‹$1.5 \times 6.0 =$› 9.0‹ V› ✓		**1**
	b	**ii**	‹X is higher› because ‹conventional› current is flowing /positive charge would flow from X to Y ✓		**1**

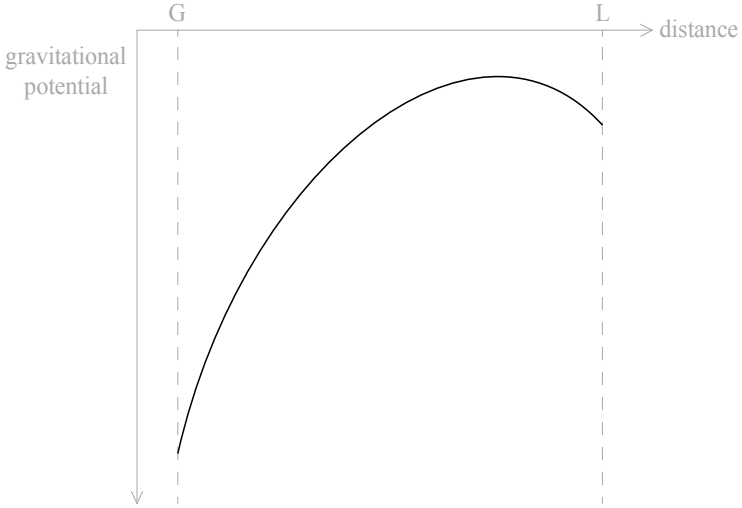
Question			Answers	Notes	Total
3.	**a**		correct curvature starting at 0,0 ✓ asymptotic to 6.0 ‹V› ✓		**2**
	b	**i**	use of $V = V_0 e^{-\frac{t}{RC}}$ ✓ $V = 6.0e^{-\frac{6.3\times10^{-3}}{3300\times4.7\times10^{-6}}}$ ✓ 4.00 ‹V› ✓		**3**
	b	**ii**	$Q_{\text{initial}} = ‹6.0\times4.7\times10^{-6} =› 28.2‹\mu C›$ ***OR*** $Q_{\text{final}} = ‹4.00\times4.7\times10^{-6} =› 18.8‹\mu C›$ ✓ $\Delta Q = 9.4‹\mu C›$ ✓	*Allow ECF from (b)(i).*	**2**
	b	**iii**	battery capacity $= 0.55\times3600 = 1980$ C ✓ $‹\frac{1980}{9.4\times10^{-6}\times3600\times24\times365} =› 6.7$ year ✓	*Allow ECF from (b)(ii).*	**2**
	b	**iv**	V values ‹all› double ✓ energy proportional to V^2 so $\times4$ ✓		**2**

Question			Answers	Notes	Total
4.	**a**		travelling sound wave ‹is reflected at the barrier and› travels in opposite direction to original wave ✓ mention of interference/superposition ✓ minima caused by destructive interference ✓ maxima caused by constructive interference ✓ ***OR*** travelling sound wave ‹reflects at the barrier› travels in opposite direction to original wave ✓ reflected wave superposes with original wave ✓ forming a standing wave ✓ maxima are positions of antinodes, minima are positions of nodes ✓		**4**
	b	**i**	recognition that 3.5 wavelengths are covered ✓ 0.29 ‹m› ✓		**2**
	b	**ii**	measure each position of several minima/maxima using a ruler ✓ use data to determine mean wavelength ✓ measure frequency of waves using *eg*: oscilloscope/frequency meter/electronic guitar tuner ✓ use of $c = f\lambda$ ✓	*Accept look up wave frequency or read from apparatus.*	**3 max**

Question			Answers	Notes	Total
5.	**a**	**i**	power $= \langle \frac{7.5\times10^{8}\times100}{38} = \rangle 1.97\times10^{9} \langle \mathrm{J\,s^{-1}} \rangle$ ✓		**1**
	a	**ii**	$\frac{1.97\times10^{9}}{35}$ ✓ 56 ✓ MJ kg^{-1} ✓		**3**
	b		plentiful supplies at present ✓ pre-existing infrastructure ✓	*Accept easily portable, easily mined.*	**2**
	c	**i**	increased proportion of greenhouse gases in atmosphere ✓ so more absorption of infrared by atmosphere ✓ and extra energy radiated back to ground ✓	*Accept any named greenhouse gas.*	**3**
	c	**ii**	improved modelling ***OR*** greater data collection ***OR*** greater international collaboration ✓		**1 max**
	d	**i**	$\langle 235+1-148-85= \rangle 3$ ✓		**1**
	d	**ii**	mass difference $= \langle 148.932+84.910+(2\times1.009)-235.044 \rangle$ ✓ $= 0.816 \langle \mathrm{u} \rangle$ ✓ 760 MeV ✓	*Allow ECF from (d)(i).*	**3**
	d	**iii**	neutrons emitted from uranium at high speed ✓ high speed neutrons do not cause fission ✓ neutrons collide with moderator atoms ✓ and therefore lose energy/speed before re-entering fuel rods ✓		**3 max**

Question			Answers	Notes	Total
6.	**a**		force of gravity is proportional to the product of ‹point› masses ✓ and inversely proportional to distance between centres squared ✓		**2**
	b	**i**	force $= \langle \frac{Gm_1m_2}{r^2} = \rangle \frac{6.67\times10^{-11}\times47\times0.73}{(1.25\times10^{-1})^2}$ ✓ 0.15 ‹μN› ✓		**2**
	b	**ii**	doubling radius increases mass of spheres by $\times 8$ ✓ so mass product increased by $\times 64$ ✓ doubling radius increases separation by factor of 2 hence r^2 by 4 ✓ overall effect is increase in force by $\times 16$ ✓		**4**
	c		same time period for any amplitude of swing ✓		**1**

Question			Answers	Notes	Total
7.	**a**		gradient calculation attempted at $t = 0$ ***or*** 0.5 ***or*** 1.0 ✓ evidence of including number of turns and area ✓ 3.2 ‹mV› ✓ answer expressed to two significant figures only ✓	*Allow 3.0 mV to 3.4 mV.*	**4**
	b		cosine or negative cosine wave ✓ peak value double answer to part a ✓ period = 0.5 ‹s› ✓		**3**
	c	**i**	 arrows as shown in diagram ✓		**1**
	c	**ii**	identifies north pole at top of coil ✓ uses stated direction rule to link induced north pole to current direction ✓	*Allow answer in terms of Lenz's law or energy considerations.*	**2**

Question			Answers	Notes	Total
8.	**a**		force acting on ✓ unit mass ✓		**2**
	b	**i**	$\frac{GM_e}{x^2} = \frac{GM_m}{y^2}$ ✓ $\frac{x}{y} = \sqrt{\frac{M_e}{M_m}} = \sqrt{81} = 9$ ✓		**2**
	b	**ii**	for the Moon $V = \langle \frac{-GM}{r} = \rangle -2.9 \langle \text{MJ kg}^{-1} \rangle$ ✓ for the Earth $\langle - \rangle 1.1 \langle \text{MJ kg}^{-1} \rangle$ ✓ total gravitational potential $= -4.0 \langle \text{MJ kg}^{-1} \rangle$ ✓		**3**
	b	**iii**	graph rises close to but below zero ✓ falls to Moon ✓ from point much closer ‹10 % of way› to Moon than Earth ✓  		**3**

Question			Answers	Notes	Total
9.	**a**		there is a minimum energy required equal to the total mass of particles created ***OR*** reference to $\Delta E = \Delta m c^2$ **and** $\Delta m = 2m_e$ ✓ energy of photon $= hf$ ‹so there is also a minimum frequency› ✓	*Both needed.*	**2**
	b		minimum energy $= 1.02$ ‹MeV› ✓ $= 1.63 \times 10^{-13}$ ‹J› ✓ 2.5×10^{20} ‹Hz› ✓		**3**
	c		‹both energy and› momentum must be conserved in the interaction ✓ nucleus absorbs the momentum ‹to allow the interaction to occur› ✓		**2**